ARQUIVISION AI

Generative-AI rendering studio for architects and interior designers.

From a client's photo to a photoreal redesign - in seconds, in their words.

LIVE - bednrixvision.ai

Final Project - Generative AI - MCSBT, IE University

Matej & Juan

THE PROBLEM

Rendering is the bottleneck in every design pitch

A studio cannot show a client what a change looks like while the conversation is still alive. One photoreal render goes through a 3D artist - and the momentum dies between meetings.

EUR 150-600

cost of a single room render

2-5 days

turnaround through a 3D artist

1 meeting

the window you have to close the sale

THE SOLUTION

Collapse days into seconds - in the client's own words

The designer uploads the client's real photo, types the change in plain Spanish or English, and gets a photoreal variation instantly - iterating live until the client says "that one."

Without ArquiVision

- Brief the 3D artist, wait days
- EUR 150-600 per room, per style
- Client decides between meetings - or not at all

With ArquiVision

- Upload the photo, type the change
- Under 30 seconds, cents per image
- Decision closed in the first meeting

Replace a EUR 150-600, multi-day task with a sub-30-second, sub-$0.30 generation.

WHO IT'S FOR

Built for the people who sell design



Architecture & interior studios

The core user: redesign a client's space live, in any style, during the meeting.



Real estate & virtual staging

Furnish and restyle empty or dated properties to sell or rent them faster.



Developers & reformers

Show buyers a finished look from a bare floor plan before a single wall is built.

THE PRODUCT

Four workspaces, one design session

Photo Enhancer

Upload a room photo, instruct edits in natural language. Style presets, masking, reference-image transfer, A/B lab.

Plan to Render

Turn a floor plan (PNG or DXF) into a furnished photoreal interior, or edit the plan with vector operations.

Video Studio

Turn any render into a cinematic architectural fly-through with the scene and its lighting preserved (Veo 3.1).

Admin Dashboard

Live usage, latency and cost telemetry per model and per day - the basis for pricing the product.

WHY IT'S NOT JUST AN API WRAPPER

The model is swappable. The intelligence around it is the product.

1

Classify intent

Spanish architectural vocabulary - cornisa, moldura, cuarteron, zocalo - mapped to a precise edit type.

>

2

Analyze scene

Read the exact camera, lighting and layout of the original, and cache it for the whole session.

>

3

Generate

Tell the model what to change and what to keep, then output a photoreal variation.



Spanish vocabulary engine



Quality tiers + fallback



Contextual edit history



Per-user cost guardrails

ARCHITECTURE

A secure proxy between the studio and the models

Frontend - React + Vite

Photo Enhancer
Plan to Render
Video Studio
Admin Dashboard
Supabase auth (JWT)

>

Backend - Flask proxy

Keys stay server-side
JWT validated per call
Tiered model routing
Daily spend cap
Usage logging

>

Model layer

OpenAI gpt-image-2
Gemini 2.5 / 3 image
Gemini 2.5 flash (vision)
Veo 3.1 (video)
Fallback chain

Supabase - Postgres + Storage + Row-Level Security

usage_logs (model, latency, cost) - creative_datasets (learning signal) - conversations & render gallery - RLS-gated inserts run as the user

TECHNICAL DEPTH

Quality tiers map to model, resolution and batch

Tier	Provider - Model	Resolution	Batch	Use
Draft	Gemini 2.5 Flash Image	default	1	Sub-3s iteration
Standard	OpenAI gpt-image-2	1K	2	Pick-from-variations
HD	OpenAI gpt-image-2	2K	2	Client-ready render
Ultra	OpenAI gpt-image-2	4K	1	Hero deliverable



Silent fallback: OpenAI to Gemini Pro to Flash on any 429/503.



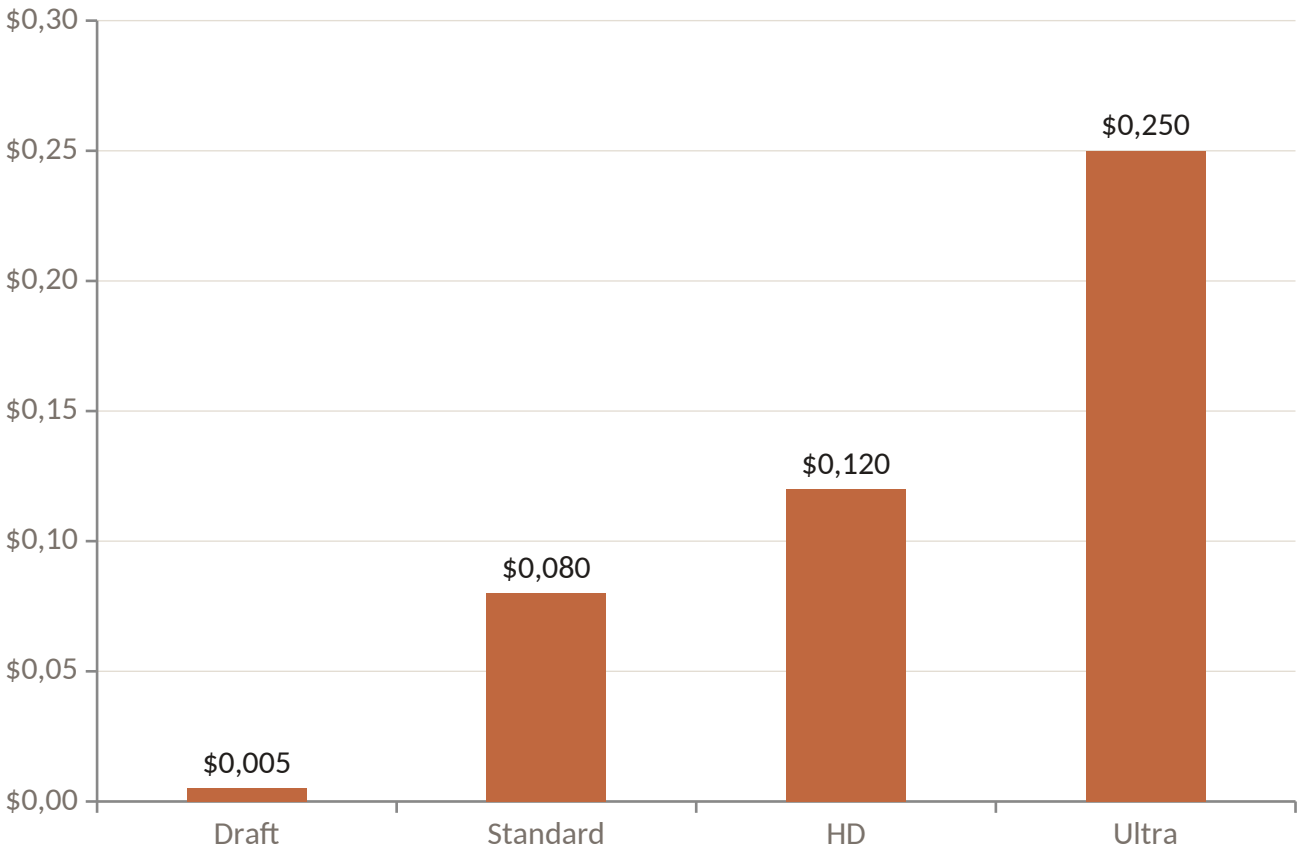
gpt-image-2 reasoning loop + native 2-image batches with object continuity.



Every call logged with model, latency and estimated cost.

UNIT ECONOMICS & GUARDRAILS

Cents per image, with a hard daily cap



vs EUR 150-600

A studio render through a 3D artist. ArquiVision delivers the same client-facing visual for a few cents.

Daily per-user spend cap

Beyond the cap, generation drops to the free Draft tier and the UI warns the user. One power user can't run up the bill.

ROADMAP

From working MVP to studio-grade product

Shipped	Next	Then	Later
gpt-image-2 swap, edit history, multi-view, Plan to Render, A/B lab	Flux + ControlNet for locked plan geometry (no wall drift)	Real upscaler for Enhance HD + a video quality tier	Multi-page client PDF deliverables & smart style suggestions

Detailed engineering plan lives in the repo: .agents/arquivision_improvement_plan.md

ArquiVision AI

Design decisions, closed in the first meeting.

Live demo: bednrixvision.ai

Matej & Juan - Generative AI - MCSBT, IE University

Thank you - questions?